Physics Basics

Physics

Lecture Notes

Physical models, measurements, and meaning.

July 14, 2026

Contents

Chapter 1

Introduction

1.1 Purpose of these notes

These notes were written for students of Computer Science who meet physics not as their main field of study, but as a part of their broader scientific and technical education. For this reason, the purpose of this script is not to reproduce a complete textbook of physics. It is not an encyclopedia, and it is not a catalogue of formulas. Its purpose is more modest, but also more important: to introduce selected parts of physics carefully enough that the reader can understand how physical reasoning works.

Physics is often presented to students as a collection of equations. In such a presentation, learning physics seems to mean remembering formulas and knowing when to substitute numbers into them. This is not the approach of these notes. A formula is not the beginning of understanding. A formula is a compressed statement about a physical model. To understand it, we must know what system is being described, what assumptions have been made, what quantities occur in the formula, what has been neglected, and in what range the formula can be used.

For example, an equation such as

$$F = ma$$

is not only a recipe for calculating force, mass, or acceleration. It expresses a relation between interaction and change of motion. It says something about the system, about the role of mass, about the meaning of acceleration, and about the way in which a simplified model of motion is being constructed. If the symbols are treated only as places where numbers should be inserted, then the physical content of the equation is lost.

One of the main goals of these notes is therefore to teach the reader how to read equations. Reading an equation does not mean pronouncing the symbols. It means understanding what the symbols represent, what kind of dependence is described, which quantities are variable and which are fixed, what happens when one quantity changes, what the limiting cases are, and whether the result makes physical sense. This skill is essential not only in physics, but also in mathematics, engineering, computer science, data analysis, simulation, modelling, and many other technical disciplines.

1.2 Mathematics as a tool for understanding

The reader should not expect these notes to avoid mathematics. Physics without mathematics would lose much of its power. At the same time, mathematics will be used here as a tool for understanding, not as a barrier. We will begin with mathematical preliminaries: functions, graphs, variables, vectors, derivatives, integrals, units, dimensions, and simple equations describing change. These topics will not be developed as an independent mathematics course. They will be introduced only to the extent needed for physical modelling.

The important point is that each mathematical object should carry meaning. A derivative is not only a formal operation; in kinematics it becomes velocity or acceleration, a rate of change of position or velocity. An integral is not only a method of finding an antiderivative; it can describe accumulation, displacement, work, or total effect. A vector is not only an ordered list of numbers; it represents a quantity with magnitude and direction. A graph is not decoration; it is information about behaviour. A differential equation is not only an equation with derivatives; it is often a compact way of saying how a system evolves.

1.3 From situation to model and back

The central method used throughout these notes can be summarized as a movement from a real or simplified situation to an idealized model, then to a mathematical description, then to calculation or simulation, and finally back to interpretation and checking.

We start from a situation in the real world or from a simplified physical scenario. Then we decide what is relevant and what can be neglected. We choose a system, variables, assumptions, and an idealized model. After that, we translate the model into mathematics. We calculate, analyse, or simulate. Finally, we return to physics: we interpret the result, check the units, consider limiting cases, and ask whether the answer makes sense.

This movement is not always linear. In real physical reasoning, we often move back and forth between meaning and mathematics. A calculation may suggest a physical interpretation. A physical idea may suggest a mathematical simplification. A unit check may reveal an error. A limiting case may show that the model is incomplete. A graph may help us understand an equation. A simulation may help us see what a formula predicts. For this reason, the notes will repeatedly return to the same pattern: from physical meaning to mathematical model, and from mathematical result back to physical meaning.

1.4 Why kinematics and dynamics matter

Special attention will be given to kinematics and dynamics. These are not only introductory chapters that must be completed before moving to more advanced topics. They are the backbone of the course. Kinematics introduces position, time, motion, velocity, acceleration, functions of time, and graphical descriptions of change. Dynamics introduces force, interaction, mass, inertia, Newton's laws, equations of motion, initial conditions, and prediction. In these chapters, the reader first encounters the full structure of physical modelling.

Later topics, such as oscillations, waves, energy, fields, or selected ideas from modern physics, should not be seen as disconnected areas. They reuse the same habits of thought. Again and again, we ask: What is the system? What are the relevant quantities? What assumptions are we making? What mathematical relation describes the model? What can be predicted? What does the result mean?

1.5 Depth rather than a catalogue of formulas

The notes deliberately prefer depth over the number of formulas. It is better to understand a small number of fundamental ideas well than to collect many equations without knowing how to use them. For each important topic, we will try to identify the physical situation, build the simplest useful model, state the assumptions, derive or motivate the mathematical description, and interpret the result. Some computations will be simple on purpose. Their role is not to test algebraic endurance, but to reveal structure.

This is especially important because many standard textbooks leave gaps in the reasoning. They often move quickly from a physical description to a formula, or from a formula to a result, assuming that the reader can reconstruct the intermediate steps. These notes should be more explicit. They should show not only what is done, but why it is done. They should make visible the transition from words to equations, from equations to consequences, and from consequences back to interpretation.

1.6 Units, dimensions, and checks

Units and dimensions will play an important role in this process. Checking units is one of the simplest and most powerful ways to test whether a formula or result can be correct. If a calculation gives a time measured in kilograms, something has gone wrong. Dimensional analysis is not a cosmetic detail; it is a basic tool of independent thinking.

Throughout the notes, the reader should learn to ask: What unit should the answer have? Do both sides of the equation have the same dimension? Does the result become larger or smaller in the expected way? What happens in extreme cases? Does the answer make physical sense?

1.7 Approximation and idealization

The notes will also emphasize the role of approximation and idealization. Physics does not describe reality by copying it in all its complexity. Instead, it builds models. We may neglect air resistance, treat a body as a point mass, assume a frictionless surface, use a small-angle approximation, or describe a complicated interaction with a simple force law. Such simplifications are not mistakes if they are made consciously. A model is useful when it captures the relevant structure of a situation and allows us to reason, calculate, predict, and understand. But every model has limits, and those limits must be remembered.

1.8 Computational experiments and simulations

The course will not rely on traditional laboratory experiments. However, experimental thinking will still be present. In many places, the reader may work with computational experiments or simulations. For Computer Science students this is natural and useful. A simple program or an AI-assisted HTML application can simulate a pendulum, projectile motion, oscillations, or waves. Such a simulation can allow us to change parameters, collect artificial measurements, estimate periods or frequencies, compare results with theory, and discuss measurement uncertainty.

A simulation is not reality. It is also a model. But precisely because it is a model, it can make the structure of physical reasoning visible. It allows us to see how assumptions become equations, how equations produce motion, how measurements are extracted, and how predictions are checked. Used correctly, computational experiments can help connect physics, mathematics, programming, and interpretation.

1.9 How to study these notes

The reader should approach these notes actively. Do not only look for final formulas. Ask what problem is being solved. Ask what assumptions are being made. Ask why a given quantity was chosen. Ask what the equation says before using it. Ask whether the answer has the right unit. Ask what would change if the assumptions changed. Ask whether the result is physically reasonable.

The goal of the course is not only to pass an exam. Passing the course is important, but it is not the deepest purpose of the notes. The deeper goal is to develop a way of thinking: to see physics as a disciplined movement between reality, idealization, mathematics, computation, and interpretation. A student who follows this path should become more comfortable with formulas, more careful with assumptions, more aware of the meaning of mathematical models, and more capable of using quantitative reasoning in future technical subjects.

In this sense, physics is not merely another subject to memorize. It is a training ground for modelling. It teaches how to take a complex situation, simplify it without losing what is essential, describe it mathematically, analyse the description, and return to the world with a prediction or explanation. This ability is valuable far beyond physics itself. It is one of the foundations of scientific and technical thinking.

Chapter 2

Mathematical Foundations

2.1 Why this chapter is necessary

This chapter collects the minimal mathematical language that will be used throughout these notes. It is not a replacement for a mathematics course. The reader is expected to have already met elementary algebra, functions, analytic geometry, vectors, matrices, derivatives, and integrals. However, experience shows that having passed a mathematics course does not always mean being able to use mathematics fluently in a new context.

The purpose of this chapter is therefore practical. We recall the mathematical objects and notations that will appear later, before they are used in more structured models. At this stage we do not develop applications in detail. Here the goal is to make the mathematical language readable.

Mathematics in these notes should not be treated as a collection of symbolic tricks. A formula is not only something into which numbers are substituted. A formula expresses a relation. A function describes dependence. A derivative describes local change. An integral describes accumulation. A vector describes magnitude and direction. A matrix can describe a transformation or organize several quantities at once.

The tools introduced here will return later with more specific meaning. For now, the goal is to understand their mathematical form.

2.2 Scientific notation

Physics and science in general often deal with very large and very small numbers. Writing such numbers in ordinary decimal notation is inconvenient and often hides the scale of the quantity. Scientific notation makes the scale visible.

Definition

A number is written in scientific notation if it has the form

$$a \times 10^n,$$

where n is an integer and usually $1 \leq |a| < 10$.

The number a carries the significant digits, while 10^n carries the scale. For example,

$$2.7 \times 10^{17} = 270000000000000000,$$

whereas

$$1.602 \times 10^{-19} = 0.0000000000000000001602.$$

The sign of the exponent tells us whether the decimal point is moved to the right or to the left.

This notation is not optional. It is a basic part of scientific language. Without it, many numbers used in science would be almost unreadable.

The elementary charge is approximately

$$e = 1.602 \times 10^{-19}\,\mathrm{C},$$

and the Avogadro constant is approximately

$$N_A = 6.022 \times 10^{23}\,\mathrm{mol}^{-1}.$$

The notation immediately shows that these quantities belong to completely different scales.

Scientific notation also makes multiplication and division easier. We use the rules

$$10^a \cdot 10^b = 10^{a+b}, \qquad \frac{10^a}{10^b} = 10^{a-b}.$$

Example

$$(2 \times 10^3)(3 \times 10^4) = 6 \times 10^{3+4} = 6 \times 10^7.$$

Also,

$$\frac{8 \times 10^6}{2 \times 10^2} = 4 \times 10^{6-2} = 4 \times 10^4.$$

Remark

The exponent of ten gives the order of magnitude. In many scientific situations, understanding the order of magnitude is more important than knowing many decimal digits.

The reader is expected to understand notation such as

$$10^{-6}, \qquad 10^3, \qquad 2.7 \times 10^{17}, \qquad 6.022 \times 10^{23}.$$

2.3 Numbers, quantities, and units

A number by itself is often not enough. In science, we usually work with quantities. A quantity has a numerical value, a unit, and a meaning.

For example, the expression

$$5$$

is only a number. It becomes a quantity only when we know what it represents:

$$5\,\mathrm{m}, \qquad 5\,\mathrm{s}, \qquad 5\,\mathrm{kg}, \qquad 5\,\mathrm{C}.$$

These expressions do not mean the same thing. The unit is part of the information.

Remark

Units are not decoration. They are part of the quantity and must be carried through a calculation consistently.

We can add quantities with the same unit:

$$2\,\mathrm{m} + 3\,\mathrm{m} = 5\,\mathrm{m}.$$

But the expression

$$2\,\mathrm{m} + 3\,\mathrm{s}$$

is not a meaningful sum, because metres and seconds describe different kinds of quantities.

Units also help us check formulas. If the left-hand side of an equation has a certain unit, then the right-hand side must have the same unit. This simple observation is one of the most useful ways to detect errors.

A calculation may be numerically correct and still be meaningless if the units do not match. Checking units does not solve the problem, but it protects us from many invalid results.

2.4 Variables, constants, and parameters

Mathematical notation uses symbols. These symbols may play different roles. Some symbols represent variables, some represent constants, and some represent parameters.

A variable is a quantity that is allowed to change. For example, in

$$f(x) = x^2 + 1,$$

the symbol x is the variable. Different values of x give different values of $f(x)$.

A constant has a fixed value. In

$$f(x) = 3x + 2,$$

the numbers 3 and 2 are constants.

A parameter is treated as fixed in one situation, but may have different values in different situations. The formula

$$f(x) = ax + b$$

describes a whole family of linear functions. For one particular function, a and b are fixed. If we choose different values of a and b, we get a different function.

Example

If $a = 2$ and $b = 5$, then

$$f(x) = 2x + 5.$$

If $a = -1$ and $b = 0$, then

$$f(x) = -x.$$

The formula is the same, but the parameters select different members of the family.

When reading a formula, ask which symbols change, which symbols remain fixed, and which symbols select the case being studied.

2.5 Functions

Definition

Let A and B be sets. A function from A to B is a rule that assigns to each element of A exactly one element of B.

The set A is the domain, and the set B is the codomain. If f is a function and x is an input from the domain, then

$$f(x)$$

denotes the output corresponding to x.

For example, let

$$f(x) = x^2 + 1.$$

Then

$$f(0) = 1, \qquad f(1) = 2, \qquad f(3) = 10.$$

Remark

The notation $f(x)$ should not be read as multiplication of f and x. It means: the value of the function f at the argument x.

A function may also depend on more than one variable. For example,

$$g(x, y) = x^2 + y^2$$

has two inputs. If $x = 3$ and $y = 4$, then

$$g(3, 4) = 3^2 + 4^2 = 25.$$

Functions are important not only because they allow us to compute values, but because they show how one quantity depends on another. The function

$$f(x) = 2x + 1$$

is linear: increasing x by 1 always increases the output by 2. The function

$$f(x) = x^2$$

is nonlinear: the change in the output depends on the current value of x.

2.6 Cartesian coordinates

To describe points on a line, in a plane, or in space, we use coordinates. The most common coordinate system is the Cartesian coordinate system.

Setting	Coordinates	Example
Line	x	2
Plane	(x, y)	$(2, 3)$
Space	(x, y, z)	$(1, -2, 5)$

This horizontal summary is often more useful than writing the three cases separately. It shows at once how the number of coordinates grows with the dimension.

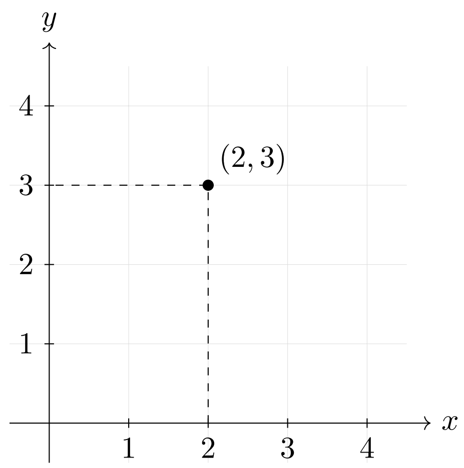

Figure 2.1: A point represented by Cartesian coordinates in the plane.

Coordinates are naturally connected with the way we describe location. A coordinate system gives a mathematical way of saying where something is.

Remark

Coordinates are not the point itself. Coordinates depend on a choice of origin, axes, orientation, and units. The same point can have different coordinates in different coordinate systems.

Geometry concerns objects such as points, lines, planes, and curves. Coordinates are a method of describing these objects using numbers.

2.7 Graphs

A graph is a visual representation of a relation. For a function

$$y = f(x),$$

we usually place x on the horizontal axis and y on the vertical axis. Each point on the graph has coordinates

$$(x, f(x)).$$

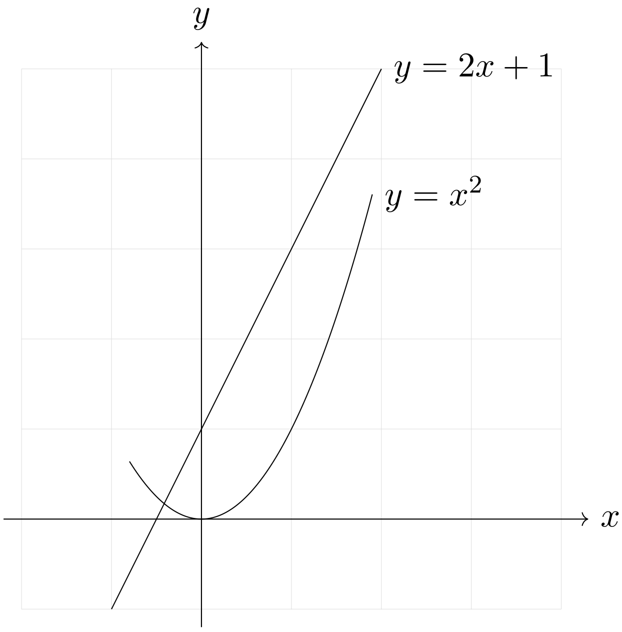

Figure 2.2: Graphs of a linear function and a quadratic function.

A graph is not only a picture. It contains information. From a graph we can often see where the function increases or decreases, where it changes rapidly or slowly, where it has zeros, and whether its shape is linear or nonlinear.

For example, the graph of $f(x) = 2x + 1$ is a straight line. The graph of $f(x) = x^2$ is a parabola.

2.8 Curves and parametric descriptions

A curve can be described in different ways. One way is to give an equation such as

$$y = f(x).$$

For example, $y = x^2$ describes a parabola in the plane.

Another way is to use a parameter. A parameter is an additional variable that describes how the coordinates are generated.

Definition

A parametric curve in the plane is a mapping

$$\mathbf{r}\colon I \to \mathbb{R}^2, \qquad \mathbf{r}(t) = (x(t), y(t)),$$

where $I \subseteq \mathbb{R}$ is an interval and $x, y\colon I \to \mathbb{R}$ are coordinate functions.

As t varies in I, the point $(x(t), y(t))$ traces the curve.

For example, the equations

$$x = t, \qquad y = t^2$$

give the same parabola as $y = x^2$, because the moving point is

$$(x, y) = (t, t^2).$$

Another important example is

$$\mathbf{r}(t) = (\cos t, \sin t).$$

This describes the unit circle, because

$$\cos^2 t + \sin^2 t = 1.$$

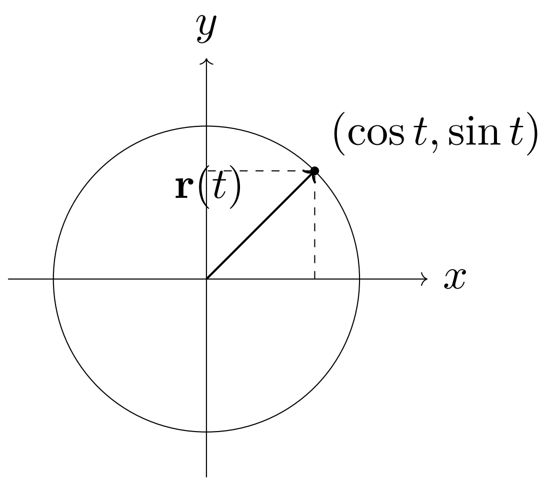

Figure 2.3: Parametric description of the unit circle.

Remark

At this stage, t is only a mathematical parameter. The important idea is that the coordinates of a moving point are functions of one parameter.

2.9 Vectors

A vector is a mathematical object with magnitude and direction. In coordinates, a vector in the plane may be written as

$$\mathbf{v} = (v_x, v_y),$$

and a vector in space as

$$\mathbf{v} = (v_x, v_y, v_z).$$

The numbers v_x, v_y, and v_z are called components. They depend on the coordinate system.

Example

The vector

$$\mathbf{v} = (3, 4)$$

has magnitude

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5.$$

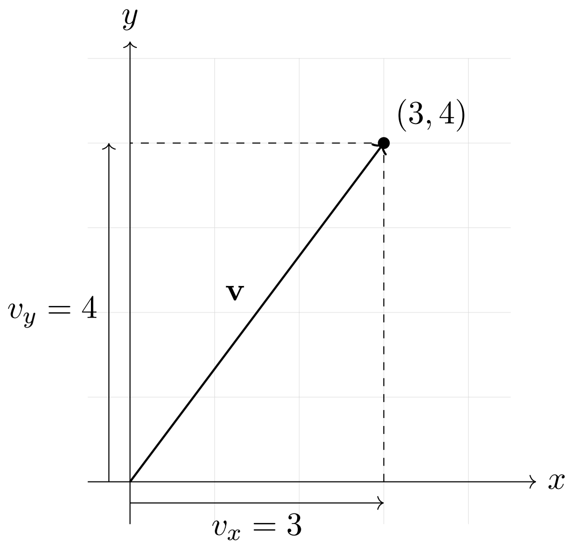

Figure 2.4: Components of a vector in the plane.

Vectors can be added component by component:

$$(1, 2) + (3, 4) = (4, 6).$$

They can be multiplied by a scalar:

$$2(1, 3) = (2, 6).$$

The zero vector is $\mathbf{0} = (0, 0)$ in the plane and $\mathbf{0} = (0, 0, 0)$ in space.

The dot product of two vectors is defined by

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$$

in two dimensions, and by

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$$

in three dimensions.

The dot product is also related to the angle between vectors:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta.$$

This formula shows that the dot product contains information about both lengths and direction.

2.10 Matrices

Definition

A matrix is a rectangular array of numbers arranged in rows and columns.

For example,

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Matrices can be added when they have the same size:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}.$$

A matrix can be multiplied by a scalar:

$$2\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

A matrix can also multiply a vector:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}.$$

Matrix multiplication is not the same as ordinary multiplication of numbers. In general,

$$AB \neq BA.$$

This means that the order of multiplication matters.

Matrices are useful because they can represent linear transformations. For example, the matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

represents a rotation in the plane by an angle θ. If

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

then $R(\theta)\mathbf{x}$ is the rotated vector.

Remark

In these notes, matrices will not be developed in detail. The reader should remember the basic idea: matrices organize several numbers and can describe transformations acting on vectors or systems of quantities.

2.11 Derivatives

The derivative describes how a function changes locally. If

$$y = f(x),$$

then the derivative of f with respect to x is written as

$$f'(x) \qquad \text{or} \qquad \frac{df}{dx}.$$

Definition

Let f be a real-valued function defined near a. If the limit

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists, then it is called the derivative of f at a and is denoted by $f'(a)$.

For example, if

$$f(x) = x^2,$$

then

$$f'(x) = 2x.$$

At $x = 1$, we get $f'(1) = 2$. At $x = 3$, we get $f'(3) = 6$. The function changes more rapidly near $x = 3$ than near $x = 1$.

If

$$f(x) = 3x + 2,$$

then

$$f'(x) = 3.$$

The derivative is constant because the function is linear and has the same slope everywhere. If $f(x) = C$, where C is constant, then $f'(x) = 0$.

The derivative also has a geometric interpretation. It is the slope of the tangent line to the graph of the function. If the derivative is positive, the function is increasing. If the derivative is negative, the function is decreasing. If the derivative is zero, the function is locally flat.

Some basic differentiation rules are

$$\frac{d}{dx} x^n = n x^{n-1}, \qquad \frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x), \qquad \frac{d}{dx}(cf(x)) = cf'(x),$$

where c is a constant.

Example

$$\frac{d}{dx}(3x^2 + 5x - 7) = 6x + 5.$$

In this chapter, the important point is purely mathematical: a derivative measures local change.

2.12 Integrals

An integral describes accumulation. It can be understood as a limiting sum or, geometrically, as the signed area under a graph. In this introductory chapter we mainly need the connection between integration and antiderivatives.

Definition

Let $f\colon I \to \mathbb{R}$ be a function defined on an interval I. A function $F\colon I \to \mathbb{R}$ is called an antiderivative of f on I if F is differentiable on I and

$$F'(x) = f(x)$$

for every $x \in I$.

Definition

The indefinite integral of f is the family of all antiderivatives of f. If F is one antiderivative of f, we write

$$\int f(x)\,dx = F(x) + C,$$

where C is an arbitrary constant.

For example,

$$\int 2x\,dx = x^2 + C,$$

because

$$\frac{d}{dx}(x^2 + C) = 2x.$$

Remark

The constant C appears because the derivative of a constant is zero. Therefore many functions can have the same derivative.

The definite integral of $f(x)$ from a to b is written as

$$\int_a^b f(x)\,dx.$$

It measures accumulation over the interval $[a, b]$. Its precise construction can be given using limiting sums; in these notes we will mainly compute it using the theorem below.

Theorem

If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Example

Since $F(x) = x^2$ is an antiderivative of $f(x) = 2x$, we have

$$\int_0^3 2x\,dx = \left[x^2\right]_0^3 = 9 - 0 = 9.$$

The symbol dx indicates the variable of integration. In the expression

$$\int_a^b f(x)\,dx,$$

we integrate with respect to x.

Some basic integration rules are

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1),$$

$$\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx, \qquad \int cf(x)\,dx = c\int f(x)\,dx.$$

Example

$$\int (3x^2 + 5x - 7)\, dx = x^3 + \frac{5}{2}x^2 - 7x + C.$$

In this chapter, the important mathematical idea is that integration describes accumulation and is closely connected with differentiation.

2.13 Differentiation and integration as inverse operations

Differentiation and integration are closely related. In a basic sense, they are inverse operations.

If

$$F'(x) = f(x),$$

then

$$\int f(x)\, dx = F(x) + C.$$

Example

Since

$$\frac{d}{dx}x^3 = 3x^2,$$

we have

$$\int 3x^2\, dx = x^3 + C.$$

For definite integrals, the connection is given by the fundamental theorem of calculus.

Theorem

If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

Example

If $f(x) = 2x$, then we can choose $F(x) = x^2$. Thus

$$\int_1^4 2x\, dx = F(4) - F(1) = 4^2 - 1^2 = 15.$$

2.14 Simple differential equations

Definition

A differential equation is an equation involving an unknown function and one or more of its derivatives.

Such an equation describes a relation between a function and its change. Unlike an algebraic equation, it does not usually ask for a single number. It asks for a function.

The equation

$$\frac{dy}{dx} = 3$$

asks for a function whose derivative is 3. The general solution is

$$y(x) = 3x + C.$$

Another simple example is

$$\frac{dy}{dx} = 2x.$$

A solution is

$$y(x) = x^2 + C,$$

because

$$\frac{d}{dx}(x^2 + C) = 2x.$$

A slightly different example is

$$\frac{dy}{dx} = ky,$$

where k is a constant. This equation says that the derivative of the function is proportional to the function itself. One family of solutions is

$$y(x) = Ce^{kx}.$$

Indeed,

$$\frac{d}{dx}Ce^{kx} = kCe^{kx} = ky(x).$$

The constant C is determined by an additional condition. For example, if

$$y(0) = 2,$$

then

$$2 = Ce^{k\cdot 0} = C,$$

so

$$y(x) = 2e^{kx}.$$

Remark

Here the important point is the basic structure: a differential equation contains derivatives and asks for an unknown function.

2.15 Approximation and limiting cases

Mathematics often uses exact expressions, but in applications we frequently use approximations. An approximation replaces a complicated expression with a simpler one under specified conditions.

For small x measured in radians,

$$\sin x \approx x.$$

Also, for small x,

$$(1 + x)^n \approx 1 + nx.$$

Remark

An approximation is useful only together with its condition of validity. Without that condition, the approximation is incomplete information.

A limiting case is a way of checking an expression by asking what happens when a variable becomes very small, very large, or takes a special value.

For example, consider

$$f(x) = \frac{x}{1+x}.$$

If x is very small, then

$$f(x) \approx x.$$

If x is very large, then

$$f(x) \approx 1.$$

These limiting behaviours help us understand the structure of the function.

Chapter 3

Kinematics

3.1 What kinematics studies

Definition

Kinematics is the part of mechanics that describes motion without explaining its causes.

At this stage we do not introduce forces, interactions, or causes of motion. Those questions belong to dynamics. In kinematics we ask a more basic question:

How can motion be described mathematically?

To answer this question, we use the mathematical language recalled in the previous chapter: coordinates, functions, parametric curves, vectors, derivatives, and integrals. The new idea is not a new mathematical object. The new idea is interpretation. We now use mathematical objects to describe the motion of a body in space and time.

The central object of kinematics is the position of a moving point as a function of time. Once position is known, we can ask how it changes. This leads to velocity. Once velocity is known, we can ask how velocity changes. This leads to acceleration:

$$\text{position} \quad \longrightarrow \quad \text{velocity} \quad \longrightarrow \quad \text{acceleration.}$$

Mathematically, this direction is built using derivatives. Later we will also move in the opposite direction:

$$\text{acceleration} \quad \longrightarrow \quad \text{velocity} \quad \longrightarrow \quad \text{position.}$$

This opposite direction is built using integrals and initial conditions.

Remark

Kinematics describes motion. Dynamics explains why motion changes. Keeping these two questions separate is essential at the beginning.

This chapter should be read slowly. The goal is not only to remember formulas, but to understand how a mathematical description of motion is constructed.

3.2 Position as a function of time

3.2.1 Position in a coordinate system

To describe motion, we must first describe position. A position is described only after we choose a coordinate system. Coordinates are therefore not the point itself; they are a numerical description of the point relative to a chosen origin, axes, orientation, and units.

Definition

In a chosen coordinate system, the position vector of a point is the vector from the origin of the coordinate system to that point.

In one dimension, position is represented by one coordinate x. In two dimensions, position may be written as (x, y), and in three dimensions as (x, y, z). It is often convenient to write position as a vector:

$$\mathbf{r} = (x, y) \qquad \text{or} \qquad \mathbf{r} = (x, y, z).$$

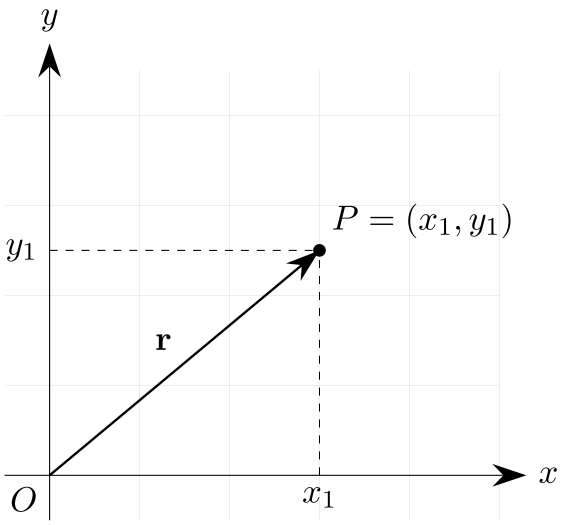

Figure 3.1: Position vector in a two-dimensional coordinate system.

Remark

If we choose a different origin or different axes, the coordinates of the same physical point may change. This is not a contradiction: coordinates depend on the chosen coordinate system.

3.2.2 Motion as changing position

A body is in motion when its position changes in time. This means that the coordinates of the point are functions of time.

Definition

A motion of a point in space is described by a position function

$$\mathbf{r} \colon I \to \mathbb{R}^3, \qquad \mathbf{r}(t) = (x(t), y(t), z(t)),$$

where $I \subseteq \mathbb{R}$ is an interval of time.

In one dimension we write $x = x(t)$. In two dimensions we write

$$\mathbf{r}(t) = (x(t), y(t)),$$

and in three dimensions

$$\mathbf{r}(t) = (x(t), y(t), z(t)).$$

The expression $\mathbf{r}(t)$ means that position depends on time. Choosing a value of t gives the position of the body at that time.

Example

If

$$\mathbf{r}(t) = (t, t^2),$$

then

$$\mathbf{r}(0) = (0,0), \qquad \mathbf{r}(1) = (1,1), \qquad \mathbf{r}(2) = (2,4).$$

One formula describes many positions: it tells us where the point is at each time.

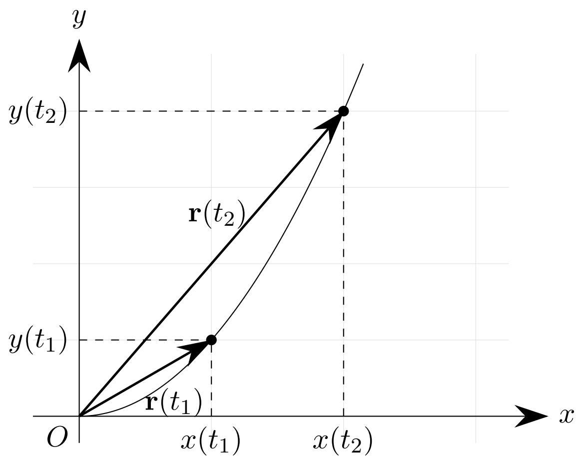

Figure 3.2: Position vectors at two different times for the motion $\mathbf{r}(t) = (t, t^2)$.

The curve shows the possible positions of the point. The labels t_1 and t_2 mark two different moments of time. At each moment, the point has a different position vector.

3.2.3 Trajectory

Definition

The trajectory of a moving point is the set of all positions occupied by that point during the motion:

$$\{\mathbf{r}(t) : t \in I\}.$$

The trajectory is a geometric curve. The motion contains more information, because it tells us where the point is at each moment of time. Two different motions may therefore have the same trajectory.

Example

The motions

$$\mathbf{r}_1(t) = (\cos t, \sin t) \qquad \text{and} \qquad \mathbf{r}_2(t) = (\cos 2t, \sin 2t)$$

both lie on the unit circle, because in both cases $x^2 + y^2 = 1$. The trajectory is the same, but the time description is different.

Geometry alone tells us the shape of the path. Kinematics needs more: it needs the position as a function of time.

3.2.4 Simple examples of position functions

Example

For motion along a straight line, one may write

$$x(t) = x_0 + at,$$

where x_0 and a are constants. If $x_0 = 1$ and $a = 2$, then

$$x(t) = 1 + 2t, \qquad x(0) = 1, \qquad x(1) = 3, \qquad x(2) = 5.$$

The formula gives the position at every time.

In two dimensions, a straight-line motion can be written as

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{a},$$

where $\mathbf{r}_0 = (x_0, y_0)$ and $\mathbf{a} = (a_x, a_y)$. In coordinates,

$$x(t) = x_0 + a_x t, \qquad y(t) = y_0 + a_y t.$$

Example

Consider

$$\mathbf{r}(t) = (t, t^2).$$

Then $x(t) = t$ and $y(t) = t^2$. Eliminating t, we get $t = x$, hence

$$y = x^2.$$

The trajectory is a parabola, while $\mathbf{r}(t) = (t, t^2)$ also contains the time description of motion along that parabola.

Another parametrization may give the same parabola with a different time description. For example,

$$\mathbf{r}(t) = (2t, 4t^2)$$

also satisfies $y = x^2$, because if $x = 2t$, then $t = x/2$, and

$$y = 4t^2 = 4\left(\frac{x}{2}\right)^2 = x^2.$$

Example

For circular motion, consider

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

where $R > 0$ is constant. Then

$$x^2(t) + y^2(t) = R^2\cos^2 t + R^2\sin^2 t = R^2.$$

Thus the trajectory is a circle of radius R centered at the origin.

The equation

$$x^2 + y^2 = R^2$$

describes the circle as a geometric object. The parametric formula $\mathbf{r}(t) = (R\cos t, R\sin t)$ describes the position on the circle at each time.

We may also write

$$\mathbf{r}(t) = (R\cos(\omega t), R\sin(\omega t)),$$

where ω is a constant. This describes the same circle, but the time dependence is different. We will return to the meaning of ω when we discuss velocity and circular motion.

3.2.5 What position tells us and what it does not tell us

A function $\mathbf{r}(t)$ is the complete kinematic description of position. It tells us where the point is at every time. From it, we can find the trajectory by looking at the set of all points $\mathbf{r}(t)$.

However, position alone is not the end of the description. Once we know how position depends on time, we naturally ask how fast the position changes and in what direction it changes. This question leads to velocity.

3.3 Velocity

3.3.1 Why position is not enough

Knowing the position of a moving point at every time tells us where the point is. But we often want a more local description: how the position is changing at a given moment.

If the position changes a lot during a short time interval, we say that the motion is fast. If the position changes only a little during the same time interval, we say that the motion is slow. If the position changes in a different direction, the motion has a different direction.

Velocity is the mathematical object that describes this change of position.

3.3.2 Average velocity

Let the position of a moving point be $\mathbf{r}(t)$. Suppose we compare two moments of time, t_1 and t_2, with $t_1 \neq t_2$. The displacement over this interval is

$$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1),$$

and the elapsed time is

$$\Delta t = t_2 - t_1.$$

Definition

The average velocity of a moving point on the interval $[t_1, t_2]$, with $t_1 \neq t_2$, is

$$\mathbf{v}_{\text{avg}} = \frac{\Delta\mathbf{r}}{\Delta t} = \frac{\mathbf{r}(t_2) - \mathbf{r}(t_1)}{t_2 - t_1}.$$

The average velocity is a vector. Its direction is the direction of the displacement $\Delta\mathbf{r}$, and its magnitude tells us how much displacement occurs per unit time during the interval. In one dimension,

$$v_{\text{avg}} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}.$$

Remark

Average velocity depends on the interval. A large interval describes overall change; a smaller interval gives a more local description.

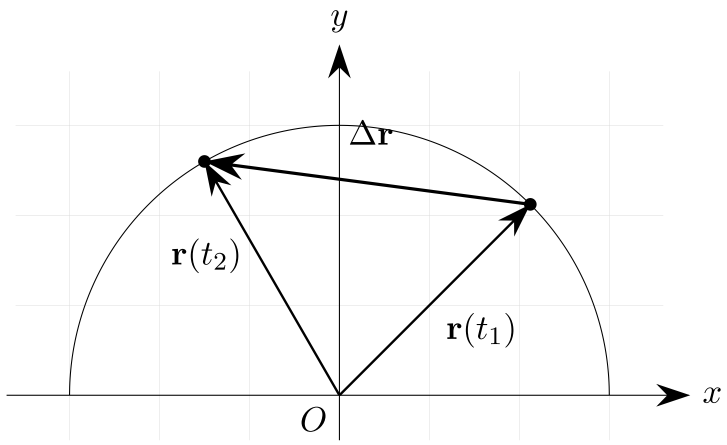

Figure 3.3: Displacement between two positions of a moving point.

3.3.3 Instantaneous velocity

Average velocity describes motion over a time interval. To describe motion at one instant, we shrink the interval.

Definition

Let $\mathbf{r}$ be differentiable at time t. The instantaneous velocity at time t is

$$\mathbf{v}(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt}(t).$$

The limiting process is essential. We are not dividing by zero. We first compute the average velocity over a nonzero time interval Δt, and only then study what this expression approaches as Δt becomes arbitrarily small.

If

$$\mathbf{r}(t) = (x(t), y(t), z(t)),$$

then

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right).$$

In two dimensions,

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}\right),$$

and in one dimension,

$$v(t) = \frac{dx}{dt}.$$

Velocity is therefore the derivative of position with respect to time.

3.3.4 Velocity as a vector attached to the moving point

Velocity is a vector quantity. It has magnitude and direction. At each time t, the point has a position $\mathbf{r}(t)$ and a velocity $\mathbf{v}(t)$. It is useful to imagine the velocity vector as attached to the point at its current position.

The velocity vector is tangent to the trajectory. The trajectory tells us the path. The velocity tells us the local direction in which the point is moving along that path.

Remark

For motion along a circle, velocity is tangent to the circle, not directed toward the center. The direction of velocity changes from point to point, even if the speed remains constant.

3.3.5 Speed

Definition

The speed of a moving point at time t is the magnitude of its velocity vector:

$$v(t) = |\mathbf{v}(t)|.$$

If

$$\mathbf{v}(t) = (v_x(t), v_y(t), v_z(t)),$$

then

$$|\mathbf{v}(t)| = \sqrt{v_x^2(t) + v_y^2(t) + v_z^2(t)}.$$

Speed is not a vector. It is a non-negative number. It tells us how fast the point is moving, but not in which direction. Velocity tells us both: how fast and in which direction.

3.3.6 Examples

Example

For one-dimensional motion

$$x(t) = x_0 + at,$$

where x_0 and a are constants, the velocity is

$$v(t) = \frac{dx}{dt} = a.$$

If $x(t) = 1 + 2t$, then $v(t) = 2$. The number 2 is the rate at which position changes with respect to time.

Example

For

$$\mathbf{r}(t) = (t, t^2),$$

we have

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}\right) = (1, 2t).$$

Thus

$$\mathbf{v}(0) = (1, 0), \qquad \mathbf{v}(1) = (1, 2), \qquad \mathbf{v}(2) = (1, 4).$$

The horizontal component is constant, while the vertical component changes.

Example

For circular motion

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

the velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (-R\sin t, R\cos t).$$

At $R = 2$ and $t = \pi/3$,

$$\mathbf{r}\left(\frac{\pi}{3}\right) = (1, \sqrt{3}), \qquad \mathbf{v}\left(\frac{\pi}{3}\right) = (-\sqrt{3}, 1).$$

Their dot product is

$$(1, \sqrt{3}) \cdot (-\sqrt{3}, 1) = 0,$$

so at this point the velocity vector is perpendicular to the position vector.

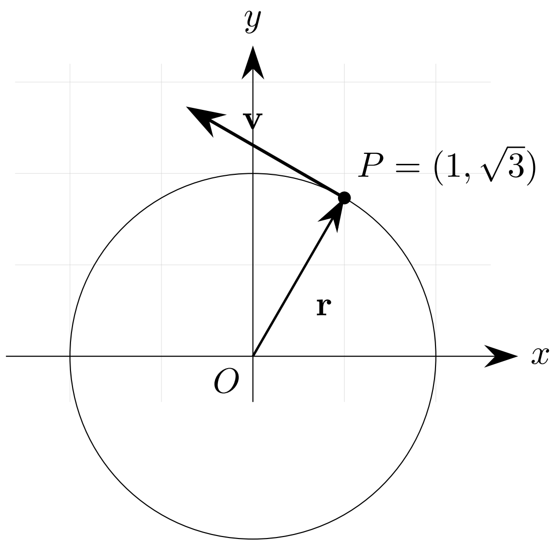

Figure 3.4: Velocity tangent to a circular trajectory.

The speed in this circular motion is

$$|\mathbf{v}(t)| = \sqrt{(-R\sin t)^2 + (R\cos t)^2} = R.$$

The speed is constant, although the velocity vector changes direction.

3.4 Acceleration

3.4.1 Why velocity is not the end of the description

Velocity describes how position changes. But velocity itself may also change. A body may move faster and faster, slow down, or move along a curve while its direction of motion changes.

Acceleration describes the change of velocity with respect to time.

3.4.2 Average acceleration

Suppose the velocity at time t_1 is $\mathbf{v}(t_1)$, and the velocity at time t_2 is $\mathbf{v}(t_2)$, with $t_1 \neq t_2$. The change in velocity is

$$\Delta\mathbf{v} = \mathbf{v}(t_2) - \mathbf{v}(t_1).$$

Definition

The average acceleration of a moving point on the interval $[t_1, t_2]$, with $t_1 \neq t_2$, is

$$\mathbf{a}_{\text{avg}} = \frac{\Delta \mathbf{v}}{\Delta t} = \frac{\mathbf{v}(t_2) - \mathbf{v}(t_1)}{t_2 - t_1}.$$

Average acceleration describes how velocity changes over a time interval.

3.4.3 Instantaneous acceleration

Definition

Let $\mathbf{v}$ be differentiable at time t. The instantaneous acceleration at time t is

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}(t).$$

Since velocity is the derivative of position,

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt},$$

we also have

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Thus acceleration is the second derivative of position with respect to time.

In coordinates,

$$\mathbf{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} \right).$$

In two dimensions,

$$\mathbf{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2} \right),$$

and in one dimension,

$$a(t) = \frac{d^2x}{dt^2}.$$

3.4.4 Acceleration is also a vector

Acceleration is a vector. It has magnitude and direction. It describes how the velocity vector changes.

Since velocity is a vector, it can change in magnitude, in direction, or in both. Therefore a body may be accelerating even if its speed is constant, provided that the direction of velocity changes.

Remark

Constant speed does not necessarily mean zero acceleration. If the direction of velocity changes, the velocity vector changes, and the acceleration is not zero.

3.4.5 Examples

Example

Consider one-dimensional motion

$$x(t) = x_0 + at,$$

where x_0 and a are constants. Then

$$v(t) = a.$$

The acceleration is

$$a_{\text{kin}}(t) = \frac{dv}{dt} = 0.$$

Here a_{kin} denotes acceleration, while a is a parameter in the formula for $x(t)$.

Example

For

$$\mathbf{r}(t) = (t, t^2),$$

we found

$$\mathbf{v}(t) = (1, 2t).$$

Therefore

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = (0, 2).$$

The acceleration is constant and points in the positive y-direction.

Example

For circular motion

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

we found

$$\mathbf{v}(t) = (-R\sin t, R\cos t).$$

Differentiating again gives

$$\mathbf{a}(t) = (-R\cos t, -R\sin t).$$

Since

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

we get

$$\mathbf{a}(t) = -\mathbf{r}(t).$$

The acceleration points toward the center of the circle.

This is a major conceptual point. In uniform circular motion, speed may be constant, but acceleration is not zero, because the direction of velocity is changing.

3.4.6 Units of acceleration

Acceleration has units of velocity divided by time. If velocity is measured in metres per second and time in seconds, acceleration is measured in

$$\mathrm{m/s^2}.$$

This also follows from

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Position contributes metres, and two differentiations with respect to time contribute two factors of seconds in the denominator. The unit $\mathrm{m/s^2}$ means metres per second per second.

3.5 Moving between position, velocity, and acceleration

3.5.1 The derivative direction

So far we have moved in one direction:

$$\mathbf{r}(t) \quad \longrightarrow \quad \mathbf{v}(t) \quad \longrightarrow \quad \mathbf{a}(t).$$

This direction uses derivatives:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}, \qquad \mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}.$$

If we know position as a function of time, we can compute velocity and acceleration by differentiating. No additional information is needed for this direction. The function $\mathbf{r}(t)$ already contains all information about the motion.

Example

If

$$\mathbf{r}(t) = (t, t^2),$$

then

$$\mathbf{v}(t) = (1, 2t), \qquad \mathbf{a}(t) = (0, 2).$$

Everything follows by differentiation.

3.5.2 The integral direction

We can also move in the opposite direction. If we know acceleration, we may want to recover velocity. If we know velocity, we may want to recover position:

$$\mathbf{a}(t) \quad \longrightarrow \quad \mathbf{v}(t) \quad \longrightarrow \quad \mathbf{r}(t).$$

This direction uses integration. Integration introduces constants or, equivalently, requires initial conditions. These are not mathematical noise; they represent missing information about the motion.

Theorem

If $\mathbf{v}(t) = d\mathbf{r}/dt$ on an interval containing $[t_0, t]$, then

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(\tau)\, d\tau.$$

If $\mathbf{a}(t) = d\mathbf{v}/dt$ on an interval containing $[t_0, t]$, then

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(\tau)\, d\tau.$$

The symbol τ is a dummy variable of integration. The upper limit t is the time at which we want to find the position or velocity. The lower limit t_0 is the initial time.

3.5.3 Why initial conditions are necessary

Knowing how something changes is not the same as knowing its value. Suppose we know only that

$$v(t) = 3.$$

Then

$$x(t) = \int 3\,dt = 3t + C.$$

The constant C is not determined by the velocity. It tells us the initial position. If $x(0) = 2$, then $C = 2$, so

$$x(t) = 3t + 2.$$

Example

Suppose acceleration is constant:

$$a(t) = A.$$

Then

$$v(t) = At + C_1.$$

If $v(0) = v_0$, then $C_1 = v_0$, so

$$v(t) = v_0 + At.$$

Integrating again gives

$$x(t) = v_0 t + \frac{1}{2}At^2 + C_2.$$

If $x(0) = x_0$, then $C_2 = x_0$, and therefore

$$x(t) = x_0 + v_0 t + \frac{1}{2}At^2.$$

This formula is often presented as a standard kinematic formula. Here it appears as the result of integrating constant acceleration twice and using initial conditions.

3.5.4 Vector form

The same idea works in vector form. If acceleration is known, then

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(\tau)\,d\tau.$$

If velocity is known, then

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(\tau)\,d\tau.$$

If acceleration is constant and equal to $\mathbf{A}$, and if $t_0 = 0$, then

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{A}t,$$

and

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{A}t^2.$$

3.5.5 The central structure of kinematics

The central structure of kinematics is

$$\boxed{\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}}$$

and

$$\boxed{\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}}.$$

The inverse relations are

$$\boxed{\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(\tau)\, d\tau}$$

and

$$\boxed{\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(\tau)\, d\tau}.$$

These formulas should not be treated as isolated recipes. They express the logic of kinematics: position describes where the body is, velocity describes how position changes, acceleration describes how velocity changes, and initial conditions supply the information needed when reconstructing motion by integration.

Dynamics will ask why acceleration appears. Before that question can be answered, we need the language developed in this chapter: position, velocity, acceleration, derivatives, integrals, and initial conditions.

Chapter 4

Dynamics

4.1 What dynamics studies

Kinematics described motion without asking for its cause. We introduced position, velocity, acceleration, derivatives, integrals, and initial conditions. The central kinematic question was:

How can motion be described mathematically?

Dynamics begins where this question becomes incomplete. Once acceleration has been defined, we naturally ask why acceleration appears and how it can be predicted.

Definition

Dynamics is the part of mechanics that studies how interactions change motion.

The central question of dynamics is therefore:

What causes acceleration?

This question should not be understood only as a search for a formula. It is a modelling question. We must decide what body is being studied, what other bodies interact with it, which interactions are relevant, which effects may be neglected, and what mathematical description follows from these choices.

In kinematics, a motion may be given directly, for example by a position function $\mathbf{r}(t)$. Then velocity and acceleration follow by differentiation:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}, \qquad \mathbf{a}(t) = \frac{d\mathbf{v}}{dt}.$$

In dynamics, the direction of reasoning is different. We usually do not start with the motion already known. Instead, we start from interactions and try to predict the acceleration. After that, the methods of kinematics allow us to reconstruct velocity and position from acceleration and initial conditions.

A schematic form of this idea is

$$\begin{aligned} \text{interactions} \longrightarrow \text{forces} \longrightarrow \text{acceleration} \\ \longrightarrow \text{motion.} \end{aligned}$$

The last arrow is already familiar from kinematics: if acceleration is known and initial conditions are given, then velocity and position can be found by integration.

4.1.1 From description to explanation

Consider a body moving along a straight line. Kinematics may tell us that its position is

$$x(t) = x_0 + v_0 t + \frac{1}{2}At^2.$$

From this expression we can find

$$v(t) = v_0 + At, \qquad a(t) = A.$$

This is a complete kinematic description of the motion. But it does not explain why the acceleration is equal to A. Dynamics asks what interaction produces this acceleration.

For example, the acceleration may be caused by gravity, by a stretched spring, by friction, by an engine, by an electric field, or by several effects at once. The same mathematical pattern of accelerated motion can therefore come from different physical models. Dynamics connects the mathematical description of acceleration with the physical interactions that produce it.

Remark

Kinematics can describe acceleration. Dynamics explains how acceleration is related to forces and interactions.

4.1.2 The system and its surroundings

The first decision in a dynamics problem is the choice of system.

Definition

The system is the body or collection of bodies whose motion we want to study. Everything outside the system is called the surroundings.

This choice is not a technical detail. Forces describe interactions between the system and other bodies, or between parts of the system. If we choose the system differently, the list of forces may change.

For example, suppose a book lies on a table. If the system is the book, then the table is part of the surroundings and exerts a contact force on the book. The Earth is also part of the surroundings and exerts the gravitational force on the book. If the system is the book together with the Earth, then the gravitational interaction is internal to the chosen system. The description changes because the system has changed.

Remark

Before writing an equation of motion, always ask: What is the system?

4.1.3 Net force as the cause of acceleration

Dynamics does not say that every force automatically produces motion in the direction of that force. Several forces may act on the same body at the same time. What matters for the acceleration of one body is the combined effect of all external forces acting on that body.

Definition

The net force acting on a body is the vector sum of all external forces acting on that body:

$$\mathbf{F}_{\text{net}} = \sum \mathbf{F}.$$

If two equal forces act in opposite directions on the same body, their sum is zero. The body may then have no acceleration, even though forces are present. Conversely, if the vector sum is not zero, the velocity of the body changes.

This is one of the most important conceptual points of dynamics. We do not ask only whether forces exist. We ask what their vector sum is.

4.1.4 Prediction and modelling

A typical dynamics problem follows a sequence of decisions:

$$\begin{aligned}\text{choose system} &\longrightarrow \text{identify interactions} \longrightarrow \text{write forces}\\ &\longrightarrow \text{sum forces} \longrightarrow \text{find acceleration.}\end{aligned}$$

After this, kinematics takes over:

$$\text{acceleration} \longrightarrow \text{velocity and position from initial conditions.}$$

The chapter will develop this sequence slowly. We will first discuss force, interaction, mass, and inertia. Then we will state Newton's laws. After that we will learn how to draw free-body diagrams and how to translate them into equations of motion.

The goal is not to collect many force formulas. The goal is to understand how a physical situation becomes a mathematical model of motion.

4.2 Force, mass, and interaction

Dynamics introduces new physical concepts that were deliberately absent from kinematics. In kinematics we could speak about acceleration without asking where it came from. In dynamics this is no longer enough. We need a language for interactions.

4.2.1 Force as a model of interaction

In elementary physics the word *force* is sometimes used as if it were an object that a body contains. This is misleading. A force is not stored inside an isolated body. A force represents an interaction between bodies.

Definition

A force is a vector quantity used to model an interaction that can change the motion of a body or deform it.

Forces have magnitude and direction. They are therefore represented by vectors. If a force acts on a body, then we must know not only how large the force is, but also in which direction it acts.

Examples of interactions include:

- the gravitational interaction between the Earth and a falling body,
- the contact interaction between a table and a book,
- the tension interaction between a rope and a body attached to it,
- the friction interaction between two surfaces,
- the elastic interaction between a spring and an attached body.

In each case, the force is not merely a name. It is a modelling decision: we decide which interaction is relevant and how it should be represented mathematically.

Remark

A force always belongs to an interaction. When naming a force, it is useful to say which body exerts it and which body receives it.

For example, instead of saying only “the normal force”, it is clearer to say “the force exerted by the table on the book”. Instead of saying only “gravity”, it is clearer to say “the gravitational force exerted by the Earth on the body”.

4.2.2 External and internal forces

The distinction between external and internal forces depends on the chosen system.

Definition

An external force on a system is a force exerted on the system by something outside the system. An internal force is a force between parts of the system.

If the system is a single block pulled by a rope, then the rope is outside the system and the rope’s pull is an external force on the block. If the system is the block together with the rope, then the interaction between the block and the rope is internal to the larger system.

For the motion of a single body, we usually write Newton’s second law using the external forces acting on that body. Internal forces become important when we study systems of many bodies, collisions, or extended objects.

4.2.3 Mass and inertia

Force alone is not enough to determine acceleration. The same net force does not produce the same acceleration for every body. A small cart is easier to accelerate than a loaded truck. The quantity that measures this resistance to acceleration is mass.

Definition

Mass is a measure of inertia: the resistance of a body to changes in its velocity.

A body with larger mass requires a larger net force to obtain the same acceleration. Equivalently, if the same net force acts on two bodies, the body with larger mass has smaller acceleration.

This meaning of mass is dynamic. It does not begin with the question “how much material is present?” but with the question “how strongly does the body resist acceleration?” In Newtonian mechanics, this inertial mass is the quantity that appears in Newton’s second law.

4.2.4 Units

In the SI system, mass is measured in kilograms:

$$[m] = \mathrm{kg}.$$

Acceleration is measured in metres per second squared:

$$[a] = \mathrm{m/s^2}.$$

Since force is related to mass times acceleration, the SI unit of force is

$$\mathrm{kg\,m/s^2}.$$

This unit is called the newton:

$$1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}.$$

Definition

One newton is the force that gives a body of mass 1 kg an acceleration of $1\,\mathrm{m/s^2}$, when it is the net force acting on that body.

This definition already suggests the form of Newton's second law. A force of 1 N is not defined as a mysterious primitive number. It is connected with the measurable effect of acceleration.

4.2.5 Forces are vectors

Because forces are vectors, they add as vectors. If two forces act along the same line and in the same direction, their magnitudes add. If they act in opposite directions, their effects partially or completely cancel. If they act in different directions, their vector sum must be found by component addition or by geometric vector addition.

In two dimensions, if

$$\mathbf{F}_1 = (F_{1x}, F_{1y}), \qquad \mathbf{F}_2 = (F_{2x}, F_{2y}),$$

then

$$\mathbf{F}_1 + \mathbf{F}_2 = (F_{1x} + F_{2x}, F_{1y} + F_{2y}).$$

For several forces,

$$\sum \mathbf{F} = \left(\sum F_x, \sum F_y\right)$$

in two dimensions, and similarly in three dimensions.

Example

Suppose two horizontal forces act on a cart. One force has magnitude 10 N to the right, and the other has magnitude 4 N to the left. If the positive direction is chosen to the right, then

$$F_{\text{net}} = 10\,\mathrm{N} - 4\,\mathrm{N} = 6\,\mathrm{N}.$$

The net force is 6 N to the right.

Example

Suppose a body is pulled by two perpendicular forces:

$$\mathbf{F}_1 = (3, 0)\,\mathrm{N}, \qquad \mathbf{F}_2 = (0, 4)\,\mathrm{N}.$$

Then

$$\mathbf{F}_{\text{net}} = (3, 4)\,\mathrm{N},$$

and its magnitude is

$$|\mathbf{F}_{\text{net}}| = \sqrt{3^2 + 4^2}\,\mathrm{N} = 5\,\mathrm{N}.$$

The net force does not point in the direction of either single force. It points in the direction of their vector sum.

4.2.6 What a force model can and cannot say

A force model describes one selected aspect of a physical situation. For example, when we write the gravitational force near Earth's surface as

$$\mathbf{F}_g = m\mathbf{g},$$

we are using an approximation: the acceleration due to gravity is treated as constant, and the height changes are assumed small compared with Earth's radius. This is a useful model, not a complete description of the Earth-body interaction.

Similarly, when we use a friction model such as

$$F_f = \mu N,$$

we are not describing every microscopic detail of two surfaces. We are using a simple macroscopic model that works reasonably well in many elementary situations, but not in all situations.

Remark

Every force formula has assumptions. A formula should be read together with the physical model that justifies it.

4.3 Newton's laws of motion

Newton's laws connect the language of motion with the language of forces. They are not isolated formulas. Together they describe how motion behaves when no net force acts, how acceleration is related to net force, and how interactions occur between pairs of bodies.

4.3.1 Newton's first law

Everyday experience may suggest that a body needs a continuous force in order to keep moving. A sliding object eventually stops, a rolling ball slows down, and a pushed box stops when the push ends. However, these examples include friction, air resistance, and other interactions. They do not show what happens when the net force is zero.

Newton's first law separates motion from the need for a sustaining force.

Theorem

Newton's first law. If the net force acting on a body is zero, then the body remains at rest or moves with constant velocity in a straight line.

In symbols, if

$$\sum \mathbf{F} = \mathbf{0},$$

then

$$\mathbf{a} = \mathbf{0},$$

and therefore

$$\mathbf{v}(t) = \mathbf{v}_0, \qquad \mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t$$

when the initial time is chosen as $t = 0$.

The law says that rest and uniform straight-line motion are dynamically equivalent. Both correspond to zero acceleration. The difference is only the value of the constant velocity.

Remark

A force is not required to maintain velocity. A nonzero net force is required to change velocity.

4.3.2 Inertial frames

Newton's first law is also used to identify a special class of reference frames.

Definition

An inertial frame of reference is a frame in which a body with zero net force moves with constant velocity.

In elementary mechanics we usually assume that the laboratory frame or the surface of the Earth is approximately inertial. This is an approximation, because the Earth rotates and moves around the Sun. For many everyday motions the approximation is sufficiently accurate.

If a frame accelerates relative to an inertial frame, Newton's laws in their simplest form do not apply directly. Additional apparent or fictitious forces may be introduced, but this belongs to a later discussion. In this chapter, unless stated otherwise, we work in an inertial frame.

4.3.3 Newton's second law

Newton's second law is the central law of dynamics. It gives the quantitative relation between net force and acceleration.

Theorem

Newton's second law. In an inertial frame, the net external force acting on a body equals the mass of the body times its acceleration:

$$\sum \mathbf{F} = m\mathbf{a}.$$

The symbol $\sum \mathbf{F}$ means the vector sum of all external forces acting on the body. It is not one selected force unless only one force acts.

If the mass is constant and nonzero, the acceleration is

$$\mathbf{a} = \frac{1}{m} \sum \mathbf{F}.$$

This equation shows two important facts:

- acceleration has the same direction as the net force,
- for a given net force, larger mass gives smaller acceleration.

In components, Newton's second law becomes separate scalar equations. In two dimensions,

$$\sum F_x = ma_x, \qquad \sum F_y = ma_y.$$

In three dimensions,

$$\sum F_x = ma_x, \qquad \sum F_y = ma_y, \qquad \sum F_z = ma_z.$$

These component equations are often the most useful form in calculations. They allow us to choose axes adapted to the problem.

Example

A body of mass 2 kg is acted on by a net horizontal force of 6 N. Newton's second law gives

$$a = \frac{F_{\text{net}}}{m} = \frac{6\,\text{N}}{2\,\text{kg}} = 3\,\text{m/s}^2.$$

The acceleration is in the direction of the net force.

4.3.4 Newton's second law as an equation of motion

The equation

$$\sum \mathbf{F} = m\mathbf{a}$$

becomes an equation of motion when acceleration is written as the second derivative of position:

$$\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2}.$$

Therefore

$$m\frac{d^2\mathbf{r}}{dt^2} = \sum \mathbf{F}.$$

This is often the most important form of Newton's second law. It connects forces directly with the unknown position function $\mathbf{r}(t)$.

If the forces depend on time, position, or velocity, then the equation may be a differential equation:

$$m\frac{d^2\mathbf{r}}{dt^2} = \mathbf{F}(t, \mathbf{r}, \mathbf{v}).$$

Solving this equation with initial conditions gives the motion.

Remark

Dynamics predicts motion by producing acceleration. Kinematics then reconstructs velocity and position from acceleration and initial conditions.

4.3.5 Newton's third law

Forces arise from interactions between bodies. Newton's third law describes the pair structure of these interactions.

Theorem

Newton's third law. If body A exerts a force on body B, then body B exerts a force on body A of equal magnitude and opposite direction.

If $\mathbf{F}_{A\to B}$ is the force exerted by A on B, and $\mathbf{F}_{B\to A}$ is the force exerted by B on A, then

$$\mathbf{F}_{A\to B} = -\mathbf{F}_{B\to A}.$$

The two forces in a third-law pair act on different bodies. For this reason, they do not cancel in the equation of motion of one body. They may cancel only when both bodies are included in the same system and we are summing forces on the whole system.

Remark

Action-reaction forces have equal magnitudes and opposite directions, but they act on different bodies. They should not be drawn as cancelling forces on a single free-body diagram.

Example

A book rests on a table. The table exerts an upward normal force on the book. The book exerts a downward force on the table. These two forces form a third-law pair, but they act on different bodies.

For the book alone, the upward normal force is balanced by the downward gravitational force exerted by the Earth. The downward force exerted by the book on the table is not a force on the book, so it does not appear in the book's equation of motion.

4.3.6 The three laws as a modelling system

The three laws should be read together:

- the first law identifies uniform motion as the natural motion when the net force is zero;
- the second law relates net force to acceleration;
- the third law reminds us that forces arise from interactions between bodies.

A typical dynamics problem therefore asks us to identify the body, identify the interactions, draw the forces acting on that body, sum those forces, and use Newton's second law to find the acceleration.

This is the point where physical reasoning becomes mathematical. We translate a situation into force vectors, translate force vectors into equations, and interpret the resulting acceleration as a prediction about motion.

4.4 Free-body diagrams

Newton's second law uses the net force acting on a body. Before we can write this net force mathematically, we must identify all forces acting on the chosen system. A free-body diagram is the standard tool for doing this carefully.

Definition

A free-body diagram is a diagram of one chosen system showing all external forces acting on that system.

The purpose of a free-body diagram is not artistic representation. Its purpose is conceptual bookkeeping. It helps us avoid two common mistakes: forgetting a force that acts on the system, or drawing a force that acts on another body instead of on the chosen system.

4.4.1 How to construct a free-body diagram

A useful procedure is the following:

1. Choose the system.
2. Represent the system by a point or a simple shape.
3. Identify all bodies in the surroundings that interact with the system.
4. Draw one force vector for each external force acting on the system.
5. Choose coordinate axes.
6. Resolve forces into components if needed.
7. Write Newton's second law in component form.

The first step is the most important. If the system is a block, draw only forces acting on the block. If the system is a cart, draw only forces acting on the cart. If the system is two connected bodies together, draw forces acting on that combined system.

Remark

A free-body diagram should contain forces acting on the chosen system, not forces exerted by the chosen system on other bodies.

4.4.2 Weight and normal force

Consider a book resting on a horizontal table. Let the system be the book. The Earth exerts a gravitational force on the book, directed downward. This force is called the weight of the book:

$$\mathbf{F}_g = m\mathbf{g}.$$

Near Earth's surface, $\mathbf{g}$ is approximately constant and directed downward, with magnitude about $9.81\,\mathrm{m/s^2}$.

The table also exerts a contact force on the book. If the surface is horizontal and there is no tendency to slide sideways, this contact force is perpendicular to the surface and directed upward. It is called the normal force and is often denoted by $\mathbf{N}$.

If the book is at rest and the vertical direction is chosen upward, Newton's second law gives

$$\sum F_y = ma_y.$$

Since the book is at rest, $a_y = 0$. Therefore

$$N - mg = 0,$$

and hence

$$N = mg.$$

This equality is not a definition of the normal force. It is the result of Newton's second law for this particular situation. In other situations, the normal force may not be equal to mg.

Remark

Do not memorize $N = mg$ as a universal rule. The normal force is whatever contact force is needed by the contact interaction, subject to the constraints of the physical situation and Newton's laws.

4.4.3 Tension

A rope, string, or cable can exert a pulling force on a body. In elementary models, a light inextensible rope transmits a force along its length. This force is called tension.

Definition

Tension is a pulling contact force transmitted by a rope, string, or cable along its direction.

If a block is pulled horizontally by a rope, the free-body diagram for the block includes the tension force exerted by the rope on the block. The direction of the tension is along the rope, away from the block.

In idealized problems, a massless rope over a frictionless pulley has the same tension throughout its length. This is a model assumption. Real ropes may have mass, elasticity, and frictional interactions with pulleys.

4.4.4 Friction

Friction is a contact interaction that opposes relative motion or the tendency of relative motion between surfaces.

There are two common elementary friction models. Kinetic friction acts when surfaces slide relative to each other. Its magnitude is often approximated by

$$F_k = \mu_k N,$$

where μ_k is the coefficient of kinetic friction and N is the magnitude of the normal force.

Static friction acts when surfaces do not slide relative to each other. Its magnitude adjusts up to a maximum value:

$$F_s \leq \mu_s N.$$

The direction of static friction opposes the tendency of relative motion.

Remark

The equation $F_s = \mu_s N$ is not generally true for static friction. The correct elementary statement is $F_s \leq \mu_s N$. Equality holds only at the threshold of slipping.

4.4.5 Choosing axes

After drawing the forces, choose coordinate axes. The axes do not have to be horizontal and vertical. They should make the equations simple.

For motion on an inclined plane, it is often convenient to choose one axis parallel to the plane and the other perpendicular to the plane. Then acceleration often has only one component along the plane, and the normal force lies along the perpendicular axis.

The force of gravity may then be decomposed into components:

$$mg \sin\theta$$

parallel to the plane, and

$$mg \cos\theta$$

perpendicular to the plane, where θ is the angle of inclination.

This decomposition is not a new physical force. It is only a mathematical rewriting of the same gravitational force in a convenient coordinate system.

4.4.6 From diagram to equations

Once the free-body diagram is complete, Newton's second law is written component by component. In two dimensions:

$$\sum F_x = ma_x, \qquad \sum F_y = ma_y.$$

The signs of the force components depend on the chosen axes. A force pointing in the positive direction contributes positively. A force pointing in the negative direction contributes negatively.

Example

A block of mass m lies on a horizontal surface and is pulled to the right by a horizontal force P. Kinetic friction of magnitude $F_k = \mu_k N$ acts to the left. The vertical forces are the normal force N upward and the weight mg downward.

If the block does not accelerate vertically, then

$$\sum F_y = 0, \qquad N - mg = 0,$$

so

$$N = mg.$$

In the horizontal direction,

$$\sum F_x = ma_x, \qquad P - F_k = ma_x.$$

Using $F_k = \mu_k N = \mu_k mg$, we get

$$a_x = \frac{P - \mu_k mg}{m}.$$

4.4.7 Common mistakes

Several mistakes appear repeatedly in dynamics problems.

First, students often draw the force that the system exerts on another body. For example, if the system is a book, the force exerted by the book on the table should not appear in the book's free-body diagram.

Second, students sometimes draw a velocity vector as if it were a force. Velocity is not a force. A body can move to the right while the net force is zero, or even while the net force points to the left.

Third, students sometimes add a separate "force of motion". There is no such force in Newtonian mechanics. Motion itself does not require a force. A force is required to change motion.

Fourth, students often assume that the normal force is always equal to mg. This is true only in certain simple cases, such as a body at rest on a horizontal surface with no other vertical forces.

Remark

A free-body diagram is a test of understanding. If the diagram is wrong, the equations that follow from it will also be wrong.

4.5 From forces to motion

The purpose of dynamics is not only to name forces. The purpose is to use forces to predict motion. This requires a connection between the physical model and the kinematic quantities introduced earlier.

The central chain of reasoning is

$$\text{physical situation} \longrightarrow \text{force model} \longrightarrow \sum \mathbf{F} = m\mathbf{a} \longrightarrow \mathbf{a}(t) \longrightarrow \mathbf{v}(t), \mathbf{r}(t).$$

This chain is the basic workflow of Newtonian mechanics.

4.5.1 The equation of motion

Newton's second law becomes an equation for motion when acceleration is replaced by the second derivative of position:

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Thus

$$m\frac{d^2\mathbf{r}}{dt^2} = \sum \mathbf{F}.$$

This equation is called an equation of motion.

Definition

An equation of motion is an equation that determines how the position of a body changes in time, usually after forces and initial conditions have been specified.

If the net force is known as a function of time, then the acceleration is known as a function of time. If the force depends on position or velocity, then the equation is a differential equation for the unknown motion.

4.5.2 Constant net force

The simplest important case occurs when the net force is constant. Suppose a body of mass m is acted on by a constant net force $\mathbf{F}_{\text{net}}$. Then Newton's second law gives

$$\mathbf{a} = \frac{\mathbf{F}_{\text{net}}}{m}.$$

The acceleration is constant. If the initial conditions at $t = 0$ are

$$\mathbf{r}(0) = \mathbf{r}_0, \qquad \mathbf{v}(0) = \mathbf{v}_0,$$

then kinematics gives

$$\mathbf{v}(t) = \mathbf{v}_0 + \frac{\mathbf{F}_{\text{net}}}{m} t,$$

and

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\frac{\mathbf{F}_{\text{net}}}{m} t^2.$$

This is not a new kinematic formula. It is the result of combining Newton's second law with the kinematics of constant acceleration.

Example

A cart of mass $m = 2\,\text{kg}$ is pulled horizontally by a constant net force of $6\,\text{N}$. The acceleration is

$$a = \frac{6\,\text{N}}{2\,\text{kg}} = 3\,\text{m/s}^2.$$

If the cart starts from rest at $x_0 = 0$, then

$$v(t) = 3t, \qquad x(t) = \frac{1}{2} \cdot 3t^2 = 1.5t^2.$$

At $t = 4\,\text{s}$,

$$v(4) = 12\,\text{m/s}, \qquad x(4) = 24\,\text{m}.$$

4.5.3 Free fall near Earth's surface

Near Earth's surface, if air resistance is neglected, the only force on a falling body is its weight:

$$\mathbf{F}_g = m\mathbf{g}.$$

Newton's second law gives

$$m\mathbf{a} = m\mathbf{g}.$$

If $m \neq 0$, then

$$\mathbf{a} = \mathbf{g}.$$

The mass cancels. In this idealized model, all bodies have the same acceleration due to gravity, independent of their masses.

If the vertical axis is chosen upward, then $g > 0$ denotes the magnitude of gravitational acceleration and

$$a_y = -g.$$

With initial conditions $y(0) = y_0$ and $v_y(0) = v_0$, we obtain

$$v_y(t) = v_0 - gt,$$

and

$$y(t) = y_0 + v_0 t - \frac{1}{2}gt^2.$$

Remark

The cancellation of mass in ideal free fall is a result of the model $\mathbf{F}_g = m\mathbf{g}$. It does not mean that mass is irrelevant in every gravitational or resistive situation.

4.5.4 Motion with friction on a horizontal surface

Consider a block of mass m sliding on a horizontal surface. Suppose a horizontal pulling force P acts to the right and kinetic friction acts to the left. If the block has no vertical acceleration, then the vertical equation gives

$$N = mg.$$

The kinetic friction magnitude is

$$F_k = \mu_k N = \mu_k mg.$$

The horizontal equation is

$$P - F_k = ma.$$

Therefore

$$a = \frac{P - \mu_k mg}{m}.$$

This expression should be interpreted carefully. If $P > \mu_k mg$, the acceleration is to the right. If $P < \mu_k mg$, the acceleration is to the left, which means the block slows down if it is moving to the right. If $P = \mu_k mg$, the acceleration is zero and the block moves with constant velocity if it was already moving.

Example

Let $m = 5\,\mathrm{kg}$, $P = 20\,\mathrm{N}$, $\mu_k = 0.2$, and $g = 10\,\mathrm{m/s^2}$. Then

$$F_k = \mu_k mg = 0.2 \cdot 5 \cdot 10 = 10\,\mathrm{N}.$$

The net horizontal force is

$$F_{\text{net}} = 20\,\mathrm{N} - 10\,\mathrm{N} = 10\,\mathrm{N},$$

so

$$a = \frac{10\,\mathrm{N}}{5\,\mathrm{kg}} = 2\,\mathrm{m/s^2}.$$

4.5.5 Spring force as a first example of a position-dependent force

Some forces are not constant. A simple and important example is the force exerted by an ideal spring. In one dimension, if x measures displacement from the equilibrium position, the spring force is modelled by Hooke's law:

$$F_s = -kx,$$

where $k > 0$ is the spring constant.

The negative sign has physical meaning. If $x > 0$, the force is negative and pulls the body back toward equilibrium. If $x < 0$, the force is positive and again pulls the body back toward equilibrium.

Newton's second law gives

$$ma = -kx.$$

Since

$$a = \frac{d^2x}{dt^2},$$

we obtain

$$m\frac{d^2x}{dt^2} = -kx.$$

Equivalently,

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0.$$

This is a differential equation. Its solution is not developed here in detail, but its form already shows an important point: when the force depends on position, dynamics naturally leads to differential equations.

Remark

A constant force gives constant acceleration. A position-dependent force usually gives an acceleration that changes with position, so the motion must be found from a differential equation.

4.5.6 A general problem-solving pattern

The following pattern should be used repeatedly in dynamics:

1. Read the physical situation and identify the body or system of interest.
2. Choose a coordinate system.
3. Draw a free-body diagram.
4. Write Newton's second law in vector or component form.
5. Substitute force models, such as mg, μN, or $-kx$, only after the forces have been identified.
6. Solve for the acceleration or for the equation of motion.
7. Use initial conditions to find velocity and position when needed.
8. Check units, signs, limiting cases, and physical meaning.

This pattern is more important than any single formula. The same pattern works for a falling body, a sliding block, a projectile, a spring, or a numerical simulation.

4.5.7 Connection with computation

For many force models, exact symbolic solutions are difficult or unnecessary. A computer can approximate the motion step by step. The idea is simple. At a given time, we know an approximate position and velocity. From the position and velocity we compute the force. From the force we compute the acceleration. Then we update velocity and position over a small time step.

A simple schematic algorithm is:

$$\mathbf{F}_n \quad \longrightarrow \quad \mathbf{a}_n = \frac{\mathbf{F}_n}{m} \quad \longrightarrow \quad \mathbf{v}_{n+1} \approx \mathbf{v}_n + \mathbf{a}_n \Delta t \quad \longrightarrow \quad \mathbf{r}_{n+1} \approx \mathbf{r}_n + \mathbf{v}_n \Delta t.$$

This is not the only numerical method, and it is not always the most accurate one. But it shows the structure of computational dynamics. The computer repeats the same physical reasoning many times: forces determine acceleration, acceleration updates velocity, and velocity updates position.

Remark

A simulation is useful only when the force model and initial conditions are understood. Computation does not replace physical reasoning; it extends it.

4.5.8 Summary of the chapter

Dynamics explains why acceleration appears. The essential relation is

$$\boxed{\sum \mathbf{F} = m\mathbf{a}}.$$

Together with the kinematic identity

$$\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2},$$

it becomes

$$\boxed{m\frac{d^2\mathbf{r}}{dt^2} = \sum \mathbf{F}}.$$

This is the central mathematical form of Newtonian dynamics.

The meaning of this equation is not only computational. It says that motion is predicted by modelling interactions, translating them into forces, summing those forces, and interpreting the resulting acceleration. Initial conditions then determine the particular motion.

Chapter 5

Oscillations and Waves

5.1 Introduction

This section provides an introduction to oscillations and waves.

Chapter 6

Electromagnetism

6.1 Introduction

This section provides an introduction to electromagnetism.

Chapter 7

Statistics and Measurements

7.1 Introduction

This section provides an introduction to statistics and measurements.

Chapter 8

Cosmology

8.1 Introduction

This section provides an introduction to cosmology.

Chapter 9

Relativity

9.1 Introduction

This section provides an introduction to relativity.

Chapter 10

Modern Physics

10.1 Introduction

This section provides an introduction to modern physics.

List of Figures